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SUMMARY:

Software Engineer with 5+ years of experience building and deploying LLM-powered applications, RAG pipelines, and agentic workflows at scale. Deep expertise in **LangChain, LangGraph, vector databases, and LLM APIs (OpenAI, Anthropic, Gemini)**. Proven track record of fine-tuning models, building multi-agent systems, and optimizing prompt engineering for accuracy, cost, and latency. Skilled in productionizing GenAI solutions with MLOps, CI/CD, and cloud infrastructure on AWS, GCP, and Azure. Passionate about building intelligent systems that solve real-world problems and push the boundaries of what AI can do.

TECHNICAL SKILLS:

- **Generative AI & LLMs:** GPT-4, GPT-4o, Claude 3.5, Gemini, LLaMA 2/3, Mistral, OpenAI APIs, Anthropic APIs, Hugging Face Transformers, LangChain, LangGraph, CrewAI, LlamaIndex
- **RAG & Vector Databases:** Pinecone, FAISS, pgvector, Chroma, Weaviate, Milvus, Embeddings (OpenAI, Cohere, Vertex), Chunking strategies, Re-ranking, Hybrid search
- **Agentic AI:** Multi-agent systems, Agent orchestration, Tool calling, Reasoning chains, ReAct, Task decomposition, MCP Servers, Autonomous agents, Human-in-the-loop
- **Prompt Engineering:** Advanced prompting, Few-shot learning, Chain-of-thought, Structured outputs, Constrained generation, Prompt optimization, Cost management
- **Fine-tuning & Model Customization:** Parameter-efficient fine-tuning (LoRA, QLoRA), Instruction tuning, RLHF, Model evaluation, Guardrails, Safety controls
- **MLOps & AI Infrastructure:** Model deployment, Model monitoring, Drift detection, CI/CD for ML, Vertex AI Pipelines, MLflow, Kubeflow, Prompt versioning, Evaluation frameworks
- **Backend & APIs:** Python (FastAPI, Django, Flask), Java, Node.js, REST APIs, GraphQL, Microservices, Async programming, Event-driven architecture
- **Cloud & Infrastructure:** AWS (Bedrock, SageMaker, EKS, Lambda), GCP (Vertex AI, Cloud Run, GKE, BigQuery), Azure (OpenAI, AKS), Docker, Kubernetes, Terraform
- **Databases:** PostgreSQL (pgvector), Redis, MongoDB, Pinecone, FAISS, Chroma, Weaviate

EDUCATION:

- **Master of Science in Computer Science**
  California State University Channel Islands, Camarillo, CA

- **Bachelor's in Computer Science**
  Gandhi Institute of Technology, Visakhapatnam, India

WORK EXPERIENCE

Blue Cross Blue Shield (via Aptek Systems), Detroit, MI
Software Engineer
Jan 2026 - Present

- Built and deployed **LLM-powered applications** using Vertex AI Gemini, LangChain, and LangGraph for claims processing and prior authorization, reducing manual document review by 30% across payer operations
- Designed and productionized **RAG pipeline** over 4M+ healthcare documents using Vertex AI Embeddings and pgvector, achieving 91% retrieval precision with sub-400ms P99 latency

- Architected **multi-agent systems** using LangGraph for healthcare workflow automation, orchestrating 4+ autonomous agents with human-in-the-loop fallbacks for clinical decision support
- Developed **advanced prompt engineering strategies** including few-shot learning, chain-of-thought, and structured output parsing optimizing for accuracy, cost, and latency
- Implemented **LLM evaluation frameworks** with golden datasets, online metrics, and drift detection tracking accuracy, faithfulness, and relevance across model versions
- Fine-tuned foundation models using **parameter-efficient techniques (LoRA)** and instruction tuning for domain-specific healthcare tasks
- Built **MLOps pipelines** with Vertex AI Model Registry, MLflow, and GitHub Actions for automated model deployment, versioning, and monitoring
- Reduced LLM API costs by 35% through prompt compression, token budgeting, response caching, and model routing logic
- Established **guardrails and safety controls** including PII/PHI detection, prompt injection prevention, and content filtering for HIPAA compliance
- Integrated **LLM APIs (OpenAI, Anthropic, Gemini)** with retry policies, structured response validation, and cost tracking across production workflows

Capital One (via Flyhigh Staffing), Warren, MI
Software Developer
Feb 2025 – Jan 2026

- Built and deployed **LLM-powered applications** using LangChain, LangGraph, and OpenAI GPT-4 for fraud detection, processing 3M+ daily transactions
- Designed **RAG pipeline** with Pinecone vector database and OpenAI embeddings over 50M+ transaction records, improving fraud signal recall by 40%
- Architected **multi-agent fraud investigation systems** using LangGraph and CrewAI, orchestrating autonomous agents with human-in-the-loop fallbacks reducing manual review by 65%
- Developed **advanced prompt engineering strategies** including chain-of-thought and tool calling for structured fraud signal extraction
- Implemented **LLM evaluation frameworks** measuring accuracy, recall, precision, and cost per inference across model versions
- Fine-tuned models for fraud classification using domain-specific data and instruction tuning
- Reduced LLM API costs by 35% through prompt compression, Redis response caching, and model routing logic
- Integrated **LLM APIs (OpenAI, Anthropic)** with retry policies, cost tracking, and token budget enforcement for production workloads
- Implemented **AI guardrails and safety controls** including PII scrubbing, content filtering, and prompt injection prevention for financial compliance

General Motors (via People Tech Group), Warren, MI
Software Engineer
Jan 2024 – Dec 2024

- Designed **RAG pipeline** with pgvector over 500k+ automotive technical manuals, reducing diagnostic lookup from 8 minutes to 45 seconds
- Architected **multi-agent diagnostic systems** using LangGraph and CrewAI, orchestrating tool-calling agents over vehicle telemetry and technical documentation
- Developed **advanced prompt engineering strategies** for domain-specific automotive applications, improving recommendation accuracy by 22%
- Implemented **LLM evaluation frameworks** tracking accuracy, faithfulness, and relevance metrics with prompt versioning
- Integrated **LLM APIs (Azure AI, Hugging Face)** with domain-specific fine-tuning and prompt optimization
- Built **MLOps pipelines** with MLflow, Prometheus, and Grafana for model monitoring, drift detection, and performance tracking

- Implemented **AI safety controls and guardrails** to ensure diagnostic recommendations met vehicle safety standards

Invicktus, Itasca, IL
Software Developer
Aug 2023 – Dec 2023

- Built and maintained production REST APIs using **Python, FastAPI, and PostgreSQL**, serving 10,000+ daily requests with P99 latency under 120ms
- Containerized services with Docker and deployed on AWS ECS with GitLab CI/CD, reducing deployment time from 4 hours to 25 minutes
- Optimized API performance through Redis caching and PostgreSQL query optimization, reducing latency by 60%
- Integrated third-party APIs for ticketing, monitoring, and alerting into a unified workflow, reducing ticket resolution time by 20%
- Designed PostgreSQL schemas for support case workflows with PgBouncer connection pooling to handle concurrent traffic spikes
- Wrote unit and integration test suites with 85%+ code coverage using pytest, catching 3 critical regressions pre-production
- Implemented API boundary data validation and error modeling in FastAPI middleware, rejecting malformed payloads early
- Documented API specs, database schemas, and deployment runbooks, enabling new engineers to ship code within their first week

California State University Channel Islands, Camarillo, CA
PMO & Web Services
Jan 2022 – May 2023

- Designed, developed, and deployed a cloud-native Parking Management System using Python, Django, React, and PostgreSQL, hosted on hybrid AWS/GCP infrastructure with 99.9% uptime and 1,000+ active users
- Engineered responsive, modular front-end components with TypeScript and React, reducing front-end issue resolution time by 30%
- Integrated OAuth 2.0, JWT, and API key authentication to secure web services, maintaining zero security incidents throughout the production lifecycle
- Optimized cloud deployment configurations using autoscaling, load balancers, and cloud function triggers, improving system responsiveness by 20% under peak loads
- Optimized real-time data ingestion and processing workflows using asynchronous task queues and event-driven services for parking alerts
- Implemented CI/CD pipelines with GitHub Actions and Docker, reducing application rollout time by 40% and enabling safe, rapid iterations
- Coordinated stakeholder feedback cycles and operational testing phases, identifying and resolving service delivery bottlenecks
- Collaborated with Infrastructure, User Services, and Solutions teams to ensure seamless service delivery and operational continuity

EzQuant, India
Software Engineer
July 2019 – Jan 2021

- Built full-stack data analytics platforms using **Python, Django, React, and PostgreSQL**, improving analytical accuracy by 25%
- Engineered large-scale data pipelines using GCP Cloud Functions, BigQuery, and MySQL (1TB+), reducing query latency by 40%
- Integrated third-party financial APIs (Yahoo Finance, RapidAPI) into backend services, enhancing data completeness by 15%

- Developed data ingestion and transformation workflows using Python for preprocessing and feature extraction
- Created interactive dashboards with React and charting libraries, increasing client engagement by 20%
- Implemented modular backend services with clean RESTful API design patterns deployed to cloud-native environments
- Practiced Agile development with 2-week sprint cycles, consistently delivering features on time with high quality
- Documented database schemas, API specifications, and data pipeline documentation

PROJECTS & OPEN SOURCE:

Job Agent AI — Full-Stack AI Job Search Platform

GitHub: github.com/yashwanth123/job-agent-ai **| Stack: Python/FastAPI · React/TypeScript · SQLAlchemy · OpenAI API · Tailwind CSS · JWT · REST APIs**

- Built an intelligent job search platform with AI-powered cover letter generation, resume tailoring, and interview preparation tools
- Designed and implemented a FastAPI backend with RESTful APIs, SQLAlchemy ORM, and JWT authentication
- Developed a React/TypeScript frontend with responsive dark theme UI and real-time API communication
- Integrated multiple job data sources (Adzuna, Remotive, Arbeitnow, Greenhouse APIs) into a unified search engine
- Created an AI content generation engine that personalizes cover letters and interview prep based on user profiles
- Implemented a smart matching algorithm scoring job relevance based on skills, preferences, and experience
- Built application tracking system with progress analytics and dashboard for monitoring job search status

Judgeval — Open-Source Contribution (Stanford/Judgment Labs)

GitHub: github.com/yashwanth123/judgeval-copy **| Stack: Python · LLM Evaluation · Agent Tracing**

- Contributed to judgeval, an open-source LLM agent evaluation and tracing framework developed by Judgment Labs (Stanford-affiliated)
- Evaluated framework issues and bugs, identifying edge cases in agent tracing and evaluation pipelines
- Tested LLM-as-a-judge metrics including tool call accuracy, hallucinations, and instruction adherence
- Implemented custom evaluators and improved multi-step evaluation workflows for agent regression testing
- Collaborated with the open-source community to submit bug reports, framework improvements, and testing enhancements
- Worked with Python-based tracing infrastructure for monitoring LLM agents in production environments